Tape Diagrams and Double Number Lines for Ratios Answer Key

Grade 6–7 Mathematics

Quick Answers

1. 5, 15
2. 12, 18, 15
3. 6, 18, 30
4. 4, 28
5. 0.8, 0.75, —
6. 4, 16, 28
7. 375, 4
8. 8, 7.5
9. 0.4, —
10. 5, 0.625

Curriculum

Level: Grade 6–7 Mathematics
6.RP.A.2 – *problems 4, 5, 8, 10*
6.RP.A.3 – *problems 1, 2, 3, 6, 7, 9, 10*
Difficulty 1–5 of 5 across 24 tagged checks.

1. Punch: juice to soda is $2:3$, with 10 cups of juice.

The juice tape is 2 boxes long and covers 10 cups, so one box is worth $10 \div 2$:
$\boxed{5}$ cups per box.
The soda tape is 3 boxes long, so it is 3×5.
$\boxed{\text{15 cups}}$

Grading note: $10 + 3 = 13$ or "add one box" means the student added instead of scaling. Every box on a tape diagram is worth the same amount, and that amount has to be found first.

2. Swimming: 4 laps in 6 minutes.

Both lines are marked in equal steps, so each step forward on the top line is one step forward on the bottom line: 4 laps and 6 minutes per step.
At 8 laps (two steps): $\boxed{12}$ minutes. At 12 laps (three steps): $\boxed{18}$ minutes.
For 10 laps, scale from the first pair: one lap takes $6 \div 4 = 1.5$ minutes, so 10 laps take 10×1.5.
$\boxed{\text{15 minutes}}$

Check: 10 laps sits between 8 and 12 laps, and 15 minutes sits between 12 and 18 minutes ✓

3. A 48 cm ribbon cut 3 : 5 red to blue.

The whole ribbon is $3 + 5 = 8$ equal boxes, so one box is $48 \div 8$:
$\boxed{6}$ cm per box.
Red is 3 boxes: 3×6.
$\boxed{18 \text{ cm}}$
Blue is 5 boxes: 5×6.
$\boxed{30 \text{ cm}}$

Check: $18 + 30 = 48$ cm, the whole ribbon ✓ *Grading note:* dividing 48 by 3 or by 5 is the classic error — the known total covers *all* the boxes, not one colour's boxes.

4. Apples: 3 kilograms cost 12 dollars.

To reach the 1 kilogram tick, divide the 3 kilogram pair by 3 — both lines by the same number:
$\boxed{4}$ dollars for one kilogram (the unit rate).
With the unit rate known, every other amount is one multiplication: 7×4.
$\boxed{28}$ dollars for 7 kilograms.

5. Pens: Brand A 5 for 4 dollars, Brand B 8 for 6 dollars.

The brands cannot be compared at 5 pens against 8 pens, so move both lines to the same place: one pen.
Brand A: $4 \div 5$ gives $\boxed{0.80}$ dollars per pen.
Brand B: $6 \div 8$ gives $\boxed{0.75}$ dollars per pen.
$\boxed{0.80 > 0.75}$
Brand B costs less per pen, so **Brand B** is the cheaper pen.

Grading note: "B is cheaper because 6 is less than 8" compares the wrong things — a price is only comparable once both are for the same number of pens.

6. Walk to ride is 4 : 7, with 12 more riders.

The ride tape is 7 boxes and the walk tape is 4 boxes, so the extra riders are the $7 - 4 = 3$ boxes that stick out. Those 3 boxes are the 12 extra students, so with x students per box:

$$3x = 12 \implies x = 12 \div 3$$

$\boxed{x = 4}$ students per box.
Walkers: 4×4.
$\boxed{16}$ students walk.
Riders: 7×4.
$\boxed{28}$ students ride.

Check: $28 - 16 = 12$ more riders ✓ (and $16 : 28$ reduces to $4 : 7$).

7. Car: 150 kilometres in 2 hours.

(a) Each step on the line is 2 hours and 150 kilometres. Five hours is 2.5 steps, so scale the pair by 2.5: 150×2.5. Equivalently the unit rate is $150 \div 2 = 75$ per hour, and 5×75 gives the same thing.
$\boxed{375}$ kilometres in five hours.
(b) 300 kilometres is exactly twice 150 kilometres, so it takes twice as long as 2 hours.
$\boxed{4}$ hours for 300 kilometres.

8. Printer: 24 pages in 3 minutes.

(a) Divide both lines by 3 to land on the 1 minute tick: $24 \div 3$.

$\boxed{8}$ pages per minute — the unit rate.

(b) Now divide the page count by the unit rate: $60 \div 8$.

$\boxed{7.5 \text{ minutes}}$

Check: $7.5 \times 8 = 60$ pages ✓. Half a minute is a legitimate answer here — the rate is steady, so the line has points between the labelled ticks.

9. Which cookie recipe is sweeter?

Recipe X is 2 boxes of sugar to 5 of flour; recipe Y is 3 to 7. Write each as sugar per part of flour:

X: $\boxed{0.4}$, since $2 \div 5 = 0.4$. Y: $3 \div 7 \approx 0.43$.

$\boxed{\frac{2}{5} < \frac{3}{7}}$

Recipe **Y** is sweeter.

Grading note: the common denominator route is just as good: $\frac{2}{5} = \frac{14}{35}$ and $\frac{3}{7} = \frac{15}{35}$. Counting boxes alone fails here — Y has more sugar boxes *and* more flour boxes.

10. Synthesis: 15 kilograms feeds 24 people.

(a) The two pairs sit on the same double number line, so they form a proportion:

$$\frac{x}{8} = \frac{15}{24} \implies 24x = 8 \times 15 = 120 \implies x = 120 \div 24$$

$\boxed{x = 5}$ kilograms feeds 8 people.

(b) The unit rate is the kilograms above 1 person: $15 \div 24$.

$\boxed{0.625}$ kilograms per person.

Check: $0.625 \times 8 = 5$ and $0.625 \times 24 = 15$ ✓ Both pairs really do lie on one line.